

UPI FRAUD INTELLIGENCE — INSIGHT REPORT

Santhosh Akshit | Python • SQL • Power BI

OVERVIEW

Analyzed 37,722 UPI transactions to identify fraud patterns and deliver prevention recommendations for a digital payments company.

TOP 5 FINDINGS

1. TIME-BASED RISK

Late Night (11PM–6AM) fraud rate: 0.45%

Morning fraud rate: 0.19%

→ Late night is 2.4x riskier than morning

2. GEOGRAPHIC CONCENTRATION

Delhi fraud rate: 0.40% (highest)

National average: 0.27%

→ Delhi is 60% above average — targeted attacks suspected

3. MERCHANT VULNERABILITY

Healthcare: 0.38% | Shopping: 0.36% | Groceries: 0.15%

→ Fraudsters prefer Healthcare — transactions seem legitimate

4. AMOUNT STRUCTURING

Small (₹500–5K): 0.50% fraud rate (highest)

Micro (<₹500): 0.27% | Large (>₹50K): 0.00%

→ Fraudsters deliberately target mid-range amounts to avoid detection

5. NEW ACCOUNT RISK

New accounts (<30 days): 0.33% fraud rate

Established accounts: 0.27% fraud rate

→ New accounts are 22% more fraud-prone

3 RECOMMENDATIONS

REC 1 — Step-up authentication after 11PM

Trigger OTP + biometric for transactions above ₹500 at night

Success metric: % reduction in late-night fraud rate MoM

REC 2 — Real-time flagging for Delhi + Healthcare/Shopping

Flag high-risk combination for manual review before settlement

Success metric: Fraud cases caught before payment clears

REC 3 — New account transaction cap

Limit to ₹2,000/day for accounts under 30 days old

Success metric: New account fraud rate reduction in 90 days

Tools: Python • SQL (SQLite) • Power BI

GitHub: [upi-fraud-analytics](#)